

TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

LAB 9

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Apache – Enable mod_userdir module

1. The Apache mod_userdir module allows visitors to access a user's website via a URL that contains that user's username such as `http://example.com/~user/` or `http://192.168.0.191/~user/`

a) Enable Apache mod_userdir in your server and write simple html code to show that the userdir already enabled. Make sure you create folder called public_html in your userdir and put the html code inside that folder.

```
> sudo a2enmod userdir
```

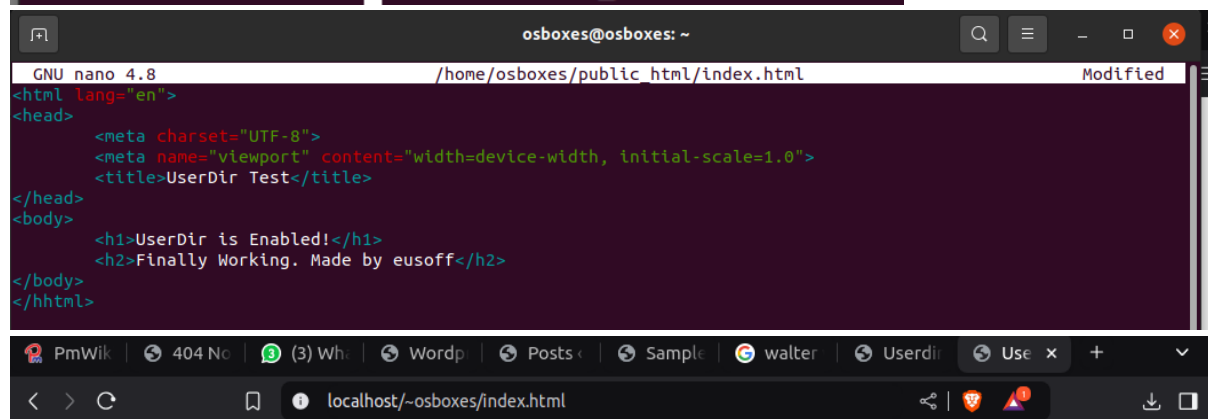
```
> sudo systemctl restart apache2
```

```
osboxes@osboxes:~$ sudo a2enmod userdir
Enabling module userdir.
To activate the new configuration, you need to run:
    systemctl restart apache2
osboxes@osboxes:~$ sudo systemctl restart apache2
```

```
> apache2ctl -M | grep userdir
```

```
osboxes@osboxes:~$ apache2ctl -M | grep userdir
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the
'ServerName' directive globally to suppress this message
userdir_module (shared)
```

```
osboxes@osboxes:~$ mkdir ~/public_html
osboxes@osboxes:~$ chmod 755 ~/public_html
osboxes@osboxes:~$ nano ~/public_html/index.html
```



```
GNU nano 4.8 /home/osboxes/public_html/index.html Modified
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>UserDir Test</title>
</head>
<body>
  <h1>UserDir is Enabled!</h1>
  <h2>Finally Working. Made by eusoff</h2>
</body>
</html>
```

UserDir is Enabled!

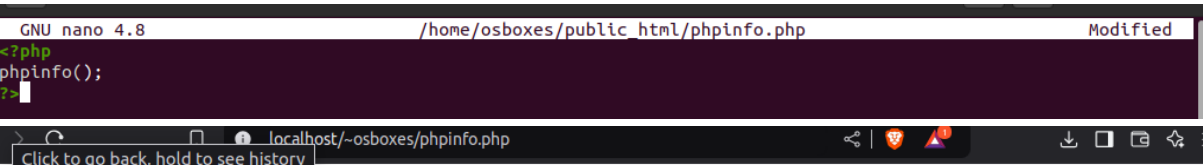
Finally Working. Made by eusoff

b) By default, PHP is disabled under mod_userdir. Edit the php configuration so that it enable PHP scripting in userdir. Make sure that PHP is enabled using phpinfo (); test code in userdir.
> sudo nano /etc/apache2/mods-enabled/php*.conf

```
# Running PHP scripts in user directories is disabled by default
#
# To re-enable PHP in user directories comment the following lines
# (from <IfModule ...> to </IfModule>.) Do NOT set it to On as it
# prevents .htaccess files from disabling it.
#<IfModule mod_userdir.c>
#   <Directory /home/*/public_html>
#       php_admin_flag engine Off
#   </Directory>
#</IfModule>
```

put hashtags

> sudo nano ~/public_html/phpinfo.php



PHP Version 7.4.3-4ubuntu2.28	
System	Linux osboxes 5.4.0-200-generic #220-Ubuntu SMP Fri Sep 27 13:19:16 UTC 2024 x86_64
Build Date	Dec 13 2024 13:46:46
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/7.4/apache2
Loaded Configuration File	/etc/php/7.4/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/7.4/apache2/conf.d
Additional .ini files parsed	/etc/php/7.4/apache2/conf.d/10-mysqld.ini, /etc/php/7.4/apache2/conf.d/10-opcache.ini, /etc/php/7.4/apache2/conf.d/10-pdo.ini, /etc/php/7.4/apache2/conf.d/20-calendar.ini, /etc/php/7.4/apache2/conf.d/20-ctype.ini, /etc/php/7.4/apache2/conf.d/20-exif.ini, /etc/php/7.4/apache2/conf.d/20-ffi.ini, /etc/php/7.4/apache2/conf.d/20-fileinfo.ini, /etc/php/7.4/apache2/conf.d/20-ftp.ini, /etc/php/7.4/apache2/conf.d/20-gettext.ini, /etc/php/7.4/apache2/conf.d/20-iconv.ini, /etc/php/7.4/apache2/conf.d/20-json.ini, /etc/php/7.4/apache2/conf.d/20-mysql.ini, /etc/php/7.4/apache2/conf.d/20-mysqli.ini, /etc/php/7.4/apache2/conf.d/20-pdo_mysql.ini, /etc/php/7.4/apache2/conf.d/20-phar.ini, /etc/php/7.4/apache2/conf.d/20-posix.ini, /etc/php/7.4/apache2/conf.d/20-readline.ini, /etc/php/7.4/apache2/conf.d/20-shmop.ini, /etc/php/7.4/apache2/conf.d/20-sockets.ini, /etc/php/7.4/apache2/conf.d/20-sysvmsg.ini, /etc/php/7.4/apache2/conf.d/20-sysvsem.ini, /etc/php/7.4/apache2/conf.d/20-sysvshm.ini, /etc/php/7.4/apache2/conf.d/20-tokenizer.ini
PHP API	20190902
PHP Extension	20190902
Zend Extension	320190902
Zend Extension Build	API320190902.NTS
PHP Extension Build	API20190902.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	disabled
IPv6 Support	enabled
DTrace Support	available, disabled
Registered PHP Streams	https, ftps, compress.zlib, php, file, glob, data, http, ftp, phar
Registered Stream Socket Transports	tcp, udp, unix, udg, ssl, tls, tlsv1.0, tlsv1.1, tlsv1.2, tlsv1.3

MySQL – Database and Web Interface

2. Web-based MySQL Database Management Clients such as phpMyAdmin or Adminer can help Sysadmin in managing online database more easily and efficiently. Install Adminer in your userdir/public_html and do the following.

> sudo apt update

> sudo apt install apache2 php mysql-server php-mysql

```
osboxes@osboxes:~$ sudo apt install apache2 php mysql-server php-mysql
Reading package lists... Done
```

> cd ~/public_html

> wget <https://github.com/vrana/adminer/releases/download/v4.8.1/adminer-4.8.1.php>

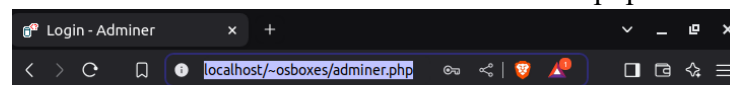
```
osboxes@osboxes:~$ cd ~/public_html
osboxes@osboxes:~/public_html$ ls
adminer  index.html  phpinfo.php
osboxes@osboxes:~/public_html$ wget https://github.com/vrana/adminer/releases/download/v4.8.1/adminer-4.8.1.php
--2025-01-16 02:21:44-- https://github.com/vrana/adminer/releases/download/v4.8.1/adminer-4.8.1.php
```

> mv adminer-4.8.1.php adminer.php

> chmod 755 adminer.php

```
osboxes@osboxes:~/public_html$ mv adminer-4.8.1.php adminer.php
osboxes@osboxes:~/public_html$ chmod 755 adminer.php
```

> Go to browser: localhost/~osboxes/adminer.php



Login

System	MySQL
Server	localhost
Username	
Password	
Database	

☐ Permanent login

Language: English

Adminer 4.8.1

System	MySQL
Server	localhost
Username	ae1207
Password	*****
Database	lab9

☐ Permanent login

a) Create database: **lab9**

> mysql -u ae1207 -p

> enter password

> SHOW DATABASES;

> CREATE DATABASE lab9;

b) Create a table called "Persons" that contains five columns: PersonID, FirstName, LastName, Address, and City.

MySQL » Server » lab9 » Create table

Language: English

Adminer 4.8.1

DB: lab9

SQL command Import Export Create table

No tables.

Create table

Table name: Persons (engine) (collation) Save

Column name	Type	Length	Options	NULL	AI?	+
PersonID	int			☐	☒	+ ↑ ↓ ×
FirstName	varchar	255	(collation)	☐	☐	+ ↑ ↓ ×
LastName	varchar	255	(collation)	☐	☐	+ ↑ ↓ ×
Address	varchar	255	(collation)	☐	☐	+ ↑ ↓ ×
City	varchar	255	(collation)	☐	☐	+ ↑ ↓ ×
	int			☐	☐	+ ↑ ↓ ×

Auto Increment: ☐ Default values ☐ Comment ☐

Save

Partition by

c) Add THREE rows of data in your table as below:

PersonID	FirstName	LastName	Address	City
1	Ali	Ahmad	12, Jalan Merak	Melaka
2	Robert	Lam	C1-2-3, Tivoli Villa	Kuala Lumpur
3	Kavita	Kaur	L12-1 Sky Apt.	Ipoh

Adminer 4.8.1

DB: lab9

[SQL command](#)
[Import](#)
[Export](#)
[Create table](#)

select Persons

Select: Persons

Item has been inserted. 02:48:26 [SQL command](#)

[Select data](#)
[Show structure](#)
[Alter table](#)
[New item](#)

Select

Search

Sort

Limit
50

Text length
100

Action
Select

SELECT * FROM `Persons` LIMIT 50 (0.001 s) [Edit](#)

☐ Modify	PersonID	FirstName	LastName	Address	City
☐ edit	1	Ali	Ahmad	12, Jalan Merak	Melaka
☐ edit	2	Robert	Lam	C1-2-3, Tivoli Villa	Kuala Lumpur
☐ edit	3	Kavita	Kaur	L12-1 Sky Apt.	Ipoh

Whole result
☐ 3 rows

Modify
Save

Selected (0)
Edit Clone Delete

Export (3)

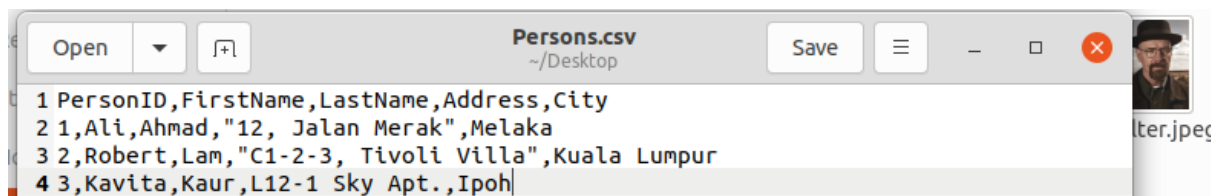
Import

d) Export and Save the data in CSV format and view the content in Web browser.

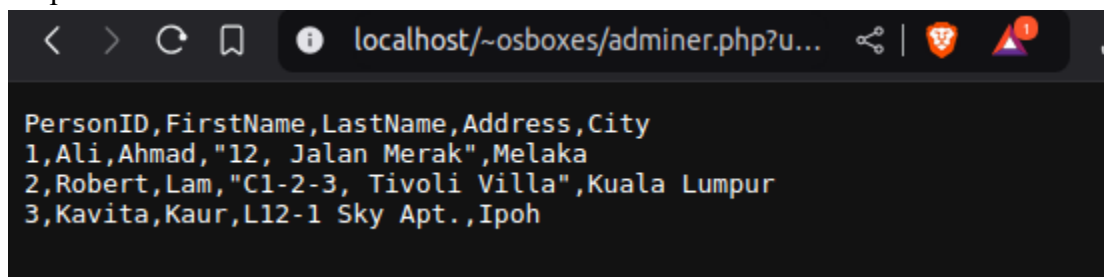
Export (3)

save
CSV
Export

> save csv



> open csv



e) Using terminal, login to MySQL, use lab9 database and show the data in Person's table.

> mysql -u ae1207 -p

> USE lab9;

> SELECT * FROM Persons;

```
mysql> USE lab9;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A
```

```
Database changed
```

```
mysql> SELECT * FROM Persons;
```

```
+-----+-----+-----+-----+-----+  
| PersonID | FirstName | LastName | Address          | City      |  
+-----+-----+-----+-----+-----+  
|      1 | Ali      | Ahmad   | 12, Jalan Merak  | Melaka    |  
|      2 | Robert   | Lam     | C1-2-3, Tivoli Villa | Kuala Lumpur |  
|      3 | Kavita   | Kaur    | L12-1 Sky Apt.   | Ipoh      |  
+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)
```